

Shaun K. Kane

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h-index 49, 10,767 citations (Google Scholar, June 2026)

EMPLOYMENT

Google, Boulder, CO

- Senior Staff Research Scientist, Human-Centered AI & Foundational Machine Learning (2026–)
- Staff Research Scientist, Human-Centered AI & Foundational Machine Learning (2022–2026)

University of Colorado Boulder, Boulder, CO

- Associate Professor Adjunct (2024–)
- Associate Professor, Department of Computer Science (2018–2024)
- Assistant Professor, Department of Computer Science (2014–2018)

University of Maryland Baltimore County, Baltimore, MD

- Assistant Professor, Department of Information Systems (2011–2014)

EDUCATION

- Ph.D. in Information Science, University of Washington, Seattle WA (2005–2011). *Co-Chairs: Jacob O. Wobbrock and Richard E. Ladner.*
- M.S. in Information Science, University of Washington, Seattle WA (2011)
- M.S. in Computer Science, University of Massachusetts, Amherst MA (2003–2005)
- B.S. in Computer Science, University of Massachusetts, Amherst MA (1999–2003)

PUBLICATIONS

Preprints / Manuscripts Under Review

- [Pr.1] Olmos, A., Sinha, A., Delos Santos, R., Rodríguez, R., Landay, J., Sepah, S., Nelson, P., and Kane, S. (2026). “Making Videos Accessible for Blind and Low Vision Users Using a Multimodal Agent Video Player”. arXiv preprint arXiv:2602.04104.

Journal Articles

- [J.12] Bennett, C., Pavel, A., Bryant, A., Eberly, L., Shelby, R., and Kane, S. (2025). ““Made by people, described by people”: The Changing Work Practices of Audio Description Professionals”. *ACM Transactions on Accessible Computing*.
- [J.11] Cai, S., Venugopalan, S., Seaver, K., Xiao, X., Tomanek, K., Jalsutram, S., Morris, M., Kane, S., Narayanan, A., MacDonald, R., Kornman, E., Vance, D., Casey, B., Gleason, S., Nelson, P., and Brenner, M. (2024). “Using large language models to accelerate communication for eye gaze typing users with ALS”. *Nature Communications*, Volume 15, Article 9449.
- [J.10] Parekh, P., Polman, J.L., Kane, S., and Shapiro, R.B. (2023). Reconfiguring science education through caring human inquiry and design with pets. *Journal of the Learning Sciences*, 32(4-5), 487-533.
- [J.9] Koushik, V. and Kane, S.K. (2022). Towards augmented reality coaching for daily routines: Participatory design with individuals with cognitive disabilities and their caregivers. *International Journal of Human-Computer Studies* 165 (2).
- [J.8] Muehlbradt, A. and Kane, S.K. (2022). What’s in an ALT tag? Exploring caption content priorities through collaborative captioning. *ACM Transactions on Accessibility* 15(1), Article 6, 32 pages.
- [J.7] Profita, H.P., Stangl, A., Matuszewska, L., Sky, S., Kushalnagar, R., and Kane, S.K. (2018). “Wear it loud”: How and why hearing aid and cochlear implant users customize their devices. *ACM Transactions on Accessibility* 11 (3), Article 13, 32 pages.
- [J.6] Wobbrock, J.O., Gajos, K.Z., Kane, S.K., and Vanderheiden, G.C. (2018). Ability-Based Design. *Communications of the ACM* 61 (6), pp. 62-71.

- [J.5] Profita, H.P., Lightner, M., Correll, N., and Kane, S.K. (2017). Textile-based wearables. *Journal on Technology and Persons with Disabilities*, 5, pp. 40-50.
- [J.4] Carrington, P., Chang, J., Chang, K., Hornback, C., Hurst, A., and Kane, S.K. (2016). The Gest-Rest Family: exploring input possibilities for wheelchair armrests. *ACM Transactions on Accessible Computing*, 8(3), article 12, 24 pages.
- [J.3] Oh, U., Branham, S., Findlater, L., and Kane, S.K. (2015). Audio-based feedback techniques for teaching touchscreen gestures. *ACM Transactions on Accessible Computing*, 7(3), article 9, 29 pages.
- [J.2] Wobbrock, J.O., Kane, S.K., Gajos, K.Z., Harada, S., Froehlich, J. (2011). Ability-Based Design: concept, principles, examples. *ACM Transactions on Accessible Computing* 3 (3), pp. 9:1-9:27.
- [J.1] Kane, S.K. (2007). Everyday inclusive web design: an activity perspective. *Information Research*, 12 (1), 16 pages.

Refereed Conference Papers

- [C.78] Li, F., Liu, M., Bennett, C., and Kane, S. (2026). “ADCanvas: Accessible and Conversational Audio Description Authoring for Blind and Low Vision Creators”. *Proceedings of CHI ‘26*.
- [C.77] Mitchell, C., Kong, J., Martinez, J., Kane, S., Ko, A., Hiniker, A., and Wobbrock, J. (2026). “Ability Heuristics for Conducting Accessibility Inspections”. *Proceedings of CHI ‘26*.
- [C.76] Bennett, C., Kane, S., and Harrington, C. (2025). “Toward Community-Led Evaluations of Text-to-Image AI Representations of Disability, Health, and Accessibility”. *Proceedings of EAAMO ‘25*, Pages 256–270.
- [C.75] Froehlich, J., Fiannaca, A., Jaber, N., Tsaran, V., and Kane, S. (2025). “StreetReaderAI: Making Street View Accessible Using Context-Aware Multimodal AI”. *Proceedings of UIST ‘25*, Article 43, Pages 1–22.
- [C.74] Moharana, S., Bennett, C., Buehler, E., Madaio, M., Tibdewal, V., and Kane, S. (2025). ““Accessibility People, You Go Work on That Thing of Yours over There”: Addressing Disability Inclusion in AI Product Organizations”. *Proceedings of AIES ‘25*, Volume 8(2), Pages 1724-1737.

- [C.73] Li, F., Wang, A., Carrington, P., and Kane, S. (2024). "A Recipe for Success? Exploring Strategies for Improving Non-Visual Access to Cooking Instructions". Proceedings of ASSETS '24, Article 26, Pages 1–15.
- [C.72] Gadiraju, V., Jayne, L., and Kane, S. (2024). "'It's an independent living skill, but covered with fun!': Prompting At-Home Skill Development for Children with Vision Impairment". Proceedings of ASSETS '24, Article 27, Pages 1–14.
- [C.71] Bennett, C., Shelby, R., Rostamzadeh, N., and Kane, S. (2024). "Painting with Cameras and Drawing with Text: AI Use in Accessible Creativity". Proceedings of ASSETS '24, Article 5, Pages 1–19.
- [C.70] Mack, K., Qadri, R., Denton, R., Kane, S., and Bennett, C. (2024). "'They only care to show us the wheelchair': disability representation in text-to-image AI models". Proceedings of CHI '24, Article 288, Pages 1–23.
- [C.69] Li, F., Liu, M., Kane, S., and Carrington, P. (2024). "A Contextual Inquiry of People with Vision Impairments in Cooking". Proceedings of CHI '24, Article 38, Pages 1–14.
- [C.68] Das, M., Fiannaca, A., Morris, M., Kane, S., and Bennett, C. (2024). "From Provenance to Aberrations: Image Creator and Screen Reader User Perspectives on Alt Text for AI-Generated Images". Proceedings of CHI '24, Article 900, Pages 1–21.
- [C.67] Gadiraju, V., Kane, S., Dev, S., Taylor, A., Wang, D., Denton, E., and Brewer, R. (2023). "'I wouldn't say offensive but...': Disability-Centered Perspectives on Large Language Models". Proceedings of FAccT '23, Pages 205–216.
- [C.66] Bowman, M., Robinson, J., Buehler, E., and Kane, S. (2023). "I just thought it was me": How smartphones fail users with mild-to-moderate dexterity differences. Proceedings of ASSETS '23, Article 4, Pages 1-12.
- [C.65] Valencia, S., Cave, R., Kallarackal, K., Seaver, K., Terry, M., and Kane, S.K. (2023). "The less I type, the better": How AI language models can enhance or impede communication for AAC users. Proceedings of CHI '23, 14 pages.
- [C.64] Wang, Q., Madaio, M., Kane, S., Kapania, S., Terry, M., and Wilcox, L. (2023). Designing responsible AI: Adaptations of UX practice to meet responsible AI challenges. Proceedings of CHI '23, 16 pages.

- [C.63] Cai, S., Venugopalan, S., Tomanek, K., Kane, S., Morris, M.R., Cave, R., MacDonald, R., Campbell, J., Casey, B., Kornman, E., Vance, D.E., and Beavers, J. (2023). SpeakFaster Observer: Long-term instrumentation of eye-gaze typing for measuring AAC communication. CHI 2023 Case Studies, 8 pages.
- [C.62] Koushik, V. and Kane, S. (2023). Ability + motivation: Understanding factors that influence people with cognitive disabilities in regularly practicing daily activities. Proceedings of W4A '23, 122-133.
- [C.61] Muehlbradt, A., Whiting, G., Kane, S., and Devendorf, L. (2022). Knitting access: Exploring stateful textiles with people with disabilities. Proceedings of DIS '22, 1058-1070.
- [C.60] Parekh, P., Polman, J.L., Kane, S.K., Shapiro, R.B. (2022). Learning science while caring for pets: Findings from a virtual workshop for teens. Proceedings of ICLS '22, 321-328.
- [C.59] Parekh, P., Polman, J.L., Kane, S.K., Shapiro, R.B. (2022). Teens and pets as participatory design partners. Proceedings of ICLS '22, 386-389.
- [C.58] Gadiraju, V., Garcia, J., Kane, S.K., and Brock, A. (2021). "It is fascinating to make these beasts fly": Understanding visually impaired people's motivations and needs for drone piloting. Proceedings of ASSETS '21, 22 pages.
- [C.57] Gadiraju, V., Doyle, O., and Kane, S.K. (2021). Exploring technology design for students with vision impairment in the classroom and remotely. Proceedings of CHI '21, Article 30, 13 pages.
- [C.56] Kane, S.K., Guo, A., and Morris, M.R. (2020). Sense and Accessibility: Understanding people with physical disabilities' experiences with sensing systems. Proceedings of ASSETS '20, Article 42, 14 pages.
- [C.55] Mott, M., Tang, J., Kane, S., Cutrell, E., and Morris, M.R. (2020). "I just went into it assuming that I wouldn't be able to have the full experience": Understanding the accessibility of virtual reality for people with limited mobility. Proceedings of ASSETS '20, Article 43, 13 pages.
- [C.54] Kelly, A., Chang, C., Hill, C., West, M., Yoder, M., Polman, J., Kane, S.K., Eisenberg, M., and Shapiro, R.B. (2020). "Our dog probably thinks Christmas is really boring": Re-mediating science education for feminist-inspired inquiry. Proceedings of ICLS '20, 935-942.

- [C.53] Gadiraju, V., Muehlbradt, A., and Kane, S.K. (2020). BrailleBlocks: Computational Braille toys for collaborative learning. Proceedings of *CHI '20*, 12 pages. (acceptance rate 23%)
- [C.52] Johnson, G.M., and Kane, S.K. (2019). Game Changer: Accessible audio and tactile guidance for board and card games. Proceedings of *W4A '20*, Article 9, 12 pages. (acceptance rate 35%)
- [C.51] Guinness, D., Muehlbradt, A., and Kane, S.K. (2019). RoboGraphics: Using mobile robots to create dynamic tactile graphics. Proceedings of *ASSETS '19*, 313-328. (acceptance rate 26%)
- [C.50] Zimmermann-Niefield, A., Turner, M., Murphy, B., Kane, S.K., and Shapiro, R.B. (2019). Youth learning machine learning through building models of athletic moves. Proceedings of *IDC '19*, 121-132. (acceptance rate 32%)
- [C.49] Reinholt, K., Guinness, D., and Kane, S.K. (2019). EyeDescribe: Combining eye gaze and speech to automatically create accessible touch screen artwork. Proceedings of *ISS '19*, 101-112. (acceptance rate 31%)
- [C.48] Koushik, V. and Kane, S.K. (2019). “It broadens my mind”: Empowering people with cognitive disabilities through computing education. Proceedings of *CHI '19*, Paper 514, 12 pages. (acceptance rate 24%)
- [C.47] Koushik, V., Guinness, D., and Kane, S.K. (2019). StoryBlocks: A tangible programming game to create accessible audio stories. Proceedings of *CHI '19*, Paper 492, 12 pages. (acceptance rate 24%)
- [C.46] Guinness, D., Muehlbradt, A., Szafir, D., and Kane, S.K. (2018). The Haptic Video Player: Using mobile robots to create tangible video annotations. Proceedings of *ISS '18*, 203-211. (acceptance rate 27%)
- [C.45] Muehlbradt, A., Atreya, M., Guinness, D., and Kane, S.K. (2018). Exploring the design of audio-kinetic graphics for education. Proceedings of *ICMI '18*, 455-463. (acceptance rate 42%)
- [C.44] Chilana, P.K., Hudson, N., Bhaduri, S., Shashikumar, P., and Kane, S.K. (2018). Supporting remote real-time help: opportunities and challenges for novice 3D modelers. Proceedings of *VL/HCC '18*, 157-166. (acceptance rate 29%)

- [C.44] Kane, S.K., Koushik, V., and Muehlbradt, A. (2018). Bonk: accessible programming for accessible audio games. Proceedings of *IDC '18*, 132–142. (acceptance rate 29%)
- [C.43] Whitlock, M., Hanner, E., Brubaker, J.R., Kane, S., and Szafir, D.A. (2018). Interacting with distant objects in augmented reality. Proceedings of *IEEE Virtual Reality Conference*, 8 pages. (acceptance rate 21%)
- [C.42] Bhaduri, S., Ortiz Tovar, J., and Kane, S.K. (2017). Fabrication Games: using 3D printers to explore new interactions for tabletop games. Proceedings of *Creativity and Cognition '17*, 51-62. (acceptance rate 29%)
- [C.41] Kane, S.K. and Morris, M.R. (2017). Let's talk about X: Combining image recognition and eye gaze to support conversation for people with ALS. Proceedings of *DIS '17*, 129-134. (acceptance rate 24%)
- [C.40] Guinness, D., Szafir, D., and Kane, S.K. (2017). GUI Robots: using off-the-shelf robots as tangible input and output devices for unmodified GUI applications. Proceedings of *DIS '17*, 767-778. (acceptance rate 24%)
- [C.39] Feit, A., Williams, S., Toledo, A., Paradiso, A., Kulkarni, H., Kane, S.K., and Morris, M.R. (2017). Toward everyday gaze input: accuracy and precision of eye tracking and implications for design. Proceedings of *CHI '17*, 1118-1130. (acceptance rate 25%).
- [C.38] Boyd, L., Rector, K., Profita, H., Stangl, A., Zolyomi, A., Kane, S.K., and Hayes, G. (2017). Understanding the role fluidity of stakeholders during assistive technology research “in the wild.” Proceedings of *CHI '17*, 6147-6158. (acceptance rate 25%)
- [C.37] Kane, S.K., Morris, M.R., Paradiso, A., and Campbell, J. (2017). “At times avuncular and cantankerous, with the reflexes of a mongoose”: understanding self-expression through augmentative and alternative communication devices. Proceedings of *CSCW '17*, 1166-1179. (acceptance rate 21%)
- [C.36] Profita, H.P., Stangl, A., Matuszewska, L., Sky, S., and Kane, S.K. (2016). Nothing to hide: aesthetic customization of hearing aids and cochlear implants in an online community. Proceedings of *ASSETS '16*, 219-227. (acceptance rate 25%)

- [C.35] Mott, M.E., Vatavu, R-D., Kane, S.K., and Wobbrock, J.O. (2016). Smart Touch: improving touch accuracy for people with motor impairments with template matching. Proceedings of *CHI '16*, 1934-1946. (acceptance rate 23%)
- [C.34] Profita, H., Albaghli, R., Findlater, L., Jaeger, P., and Kane, S.K. (2016). The AT Effect: how disability affects the perceived social acceptability of wearable computing use. Proceedings of *CHI '16*, 4884-4895. (acceptance rate 23%)
- [C.33] Morris, M.R., Perkins, A.Z., Yao, C., Bahram, S., Bigham, J.P., and Kane, S.K. (2016). “With most of it being pictures now, I rarely use it”: understanding twitter’s evolving accessibility to blind users. Proceedings of *CHI '16*, 5506-5516. (acceptance rate 23%)
- [C.32] Branham, S., and Kane, S.K. (2015). The invisible work of accessibility: how blind employees manage accessibility in mixed-ability workplaces. Proceedings of *ASSETS '15*, 163-171. (acceptance rate 23%)
- [C.31] Williams, M.A., Buehler, E., Hurst, A., and Kane, S.K. (2015). What not to wearable: using participatory workshops to explore wearable device form factors for blind users. Proceedings of *W4A '15*, article 31. (acceptance rate 35%)
- [C.30] Carrington, P.A., Hosmer, S., Yeh, T., Hurst, A., and Kane, S.K. (2015). “Like this, but better”: supporting novices’ design and fabrication of 3D models using existing objects. Proceedings of *iConference '15*. (acceptance rate 36%)
- [C.29] Branham, S. and Kane, S.K. (2015). Collaborative accessibility: how blind and sighted companions co-create accessible home spaces. Proceedings of *CHI '15*, 2373-2382. (acceptance rate 25%)
- [C.28] Buehler, E., Branham, S., Ali, A., Chang, J., Hofmann, M., Hurst, A., and Kane, S.K. (2015). Sharing is caring: assistive technology designs on Thingiverse. Proceedings of *CHI '15*, 525-534. (acceptance rate 25%).
- [C.27] Zyskowski, K., Morris, M.R., Bigham, J.P., Gray, M., and Kane, S.K. (2015). Accessible crowdwork? Understanding the value in and challenge of microtask employment for people with disabilities. Proceedings of *CSCW '15*, 1682-1693. (acceptance rate 28%)

- [C.26] Shewbridge, R., Hurst, A., Kane, S.K. (2014). Everyday making: identifying future uses for 3D printing in the home. Proceedings of *DIS '14*, 815-824. (acceptance rate 27%)
- [C.25] Carrington, P., Hurst, A., and Kane, S.K. (2014). The Gest-Rest: a pressure-sensitive chairable input pad for power wheelchair armrests. Proceedings of *ASSETS '14*, 201-208. (acceptance rate 26%)
- [C.24] Williams, M.A., Galbraith, C., Kane, S.K., and Hurst, A. (2014). “Just let the cane hit it”: how the blind and sighted see navigation differently. Proceedings of *ASSETS '14*, 217-224. (acceptance rate 26%)
- [C.23] Buehler, E., Kane, S.K., and Hurst, A. (2014). ABC and 3D: opportunities and obstacles to 3D printing in special education environments. Proceedings of *ASSETS '14*, 107-114. (acceptance rate 26%)
- [C.22] Carrington, P., Hurst, A., and Kane, S.K. (2014). Wearables and chairables: inclusive design of mobile input and output devices for power wheelchair users. Proceedings of *CHI '14*, 3101-3112. (acceptance rate 27%).
- [C.21] Kane, S.K. and Bigham, J.P. (2014). Tracking @stemxcomet: teaching programming to blind students via 3D printing, crisis management, and Twitter. Proceedings of *SIGCSE '14*, 247-252. (acceptance rate 39%)
- [C.20] Said, K., Williams, M.A., Hurst, A., and Kane, S.K. (2014). Framing the conversation: the role of Facebook conversations in shopping for eyeglasses. Proceedings of *CSCW '14*, 652-661. (acceptance rate 27%)
- [C.19] Kane, S.K., Morris, M.R., and Wobbrock, J.O. (2013). Touchplates: low-cost tactile overlays for visually impaired touch screen users. Proceedings of *ASSETS '13*, article 22, 8 pages. (acceptance rate 29%)
- [C.18] Williams, M.A., Hurst, A.K., and Kane, S.K. (2013). “Pray before you step out”: describing personal and situational blind navigation behaviors. Proceedings of *ASSETS '13*, article 28, 8 pages. (acceptance rate 29%)
- [C.17] Oh, U., Kane, S.K., and Findlater, L. (2013). Follow that sound: using sonification and corrective verbal feedback to teach touchscreen gestures. Proceedings of *ASSETS '13*, article 13, 8 pages. (acceptance rate 29%)
- [C.16] Kane, S.K., Frey, B., and Wobbrock, J.O. (2013). Access Lens: a gesture-based screen reader for real-world documents. Proceedings of *CHI '13*, 347-350. (acceptance rate 20%)

- [C.15] Kane, S.K., Linam-Church, B., Althoff, K., and McCall, D. (2012). What we talk about: designing a context-aware communication tool for people with aphasia. *Proceedings of ASSETS '12*, 49-56. (acceptance rate 28%)
- [C.14] Kane, S.K., Morris, M.R., Perkins, A.Z., Wigdor, D., Ladner, R.E., and Wobbrock, J.O. (2011). Access Overlays: improving non-visual access to large touch screens for blind users. *Proceedings of UIST '11*, 273-282. (acceptance rate 26%)
- [C.13] Kane, S.K., Wobbrock, J.O., and Ladner, R.E. (2011). Usable gestures for blind people: understanding preference and performance. *Proceedings of CHI '11*, 413-422. (acceptance rate 26%)
- [C.12] Rosenthal, S., Kane, S.K., Wobbrock, J.O., and Avrahami, D. (2010). Augmenting on-screen instructions with micro-projected guides: when it works, and when it fails. *Proceedings of Ubicomp '10*, 203-212. (acceptance rate 19%)
- [C.11] Kane, S.K., Avrahami, D., Wobbrock, J.O., Harrison, B., Rea, A.D., Philipose, M., and LaMarca, A. (2009). Bonfire: a nomadic system for hybrid laptop-tabletop interaction. *Proceedings of UIST '09*, 129-138. (acceptance rate 19%)
- [C.10] Kane, S.K., Jayant, C., Wobbrock, J.O., and Ladner, R.E. (2009). Freedom to roam: a study of mobile device adoption and accessibility for people with visual and motor disabilities. *Proceedings of ASSETS '09*, 115-122. (acceptance rate 31%)
- [C.9] Kane, S.K., Karlson, A.K., Meyers, B.R., Johns, P., Jacobs, A., and Smith, G. (2009). Exploring cross-device web use on PCs and mobile devices. *Proceedings of INTERACT '09*, 722-735. (acceptance rate 29%)
- [C.8] Karlson, A.K., Meyers, B.R., Jacobs, A., Johns, P., and Kane, S.K. (2009). Working overtime: patterns of smartphone and PC usage in the day of an information worker. *Proceedings of Pervasive '09*, 398-405. (acceptance rate 21%)
- [C.7] Kane, S.K., Bigam, J.P., and Wobbrock, J.O. (2008). Slide Rule: Making mobile touch screens accessible to blind people using multi-touch interaction techniques. *Proceedings of ASSETS '08*, 73-80. (acceptance rate 37%)

- [C.6] Kane, S.K., Wobbrock, J.O., and Smith, I.E. (2008). Getting off the treadmill: evaluating walking user interfaces for mobile devices in public spaces. Proceedings of *MobileHCI '08*, 109-118. (acceptance rate 32%)
- [C.5] Kane, S.K., Wobbrock, J.O., Harniss, M., and Johnson, K.L. (2008). TrueKeys: identifying and correcting typing errors for people with motor impairments. Proceedings of *IUI '08*, 349-352. (acceptance rate 31%)
- [C.4] Kahn, P.H., Jr., Freier, N.G., Kanda, T., Ishiguro, H., Ruckert, J.H., Severson, R.L., and Kane, S.K. (2008). Design patterns for sociality in human-robot interaction. Proceedings of *HRI '08*, ACM, 97-104. (acceptance rate 18%)
- [C.3] Nathan, L.P., Friedman, B., Klasnja, P., Kane, S.K., and Miller, J.K. (2008). Envisioning systemic effects on persons and society throughout interactive system design. Proceedings of *DIS '08*, 1-10. (acceptance rate 34%)
- [C.2] Froehlich, J., Wobbrock, J.O., and Kane, S.K. (2007). Barrier Pointing: Using physical edges to assist target acquisition on mobile device touch screens. Proceedings of *ASSETS '07*, 19-26. (acceptance rate 31%)
- [C.1] Kane, S.K., Shulman, J.A., Shockley, T.J., and Ladner, R. E. (2007). A web accessibility report card for top university web sites. Proceedings of *W4A '07*, 148-156. (acceptance rate 40%)

Refereed Posters, Extended Abstracts, and Presentations

- [P.22] Froehlich, J., Fiannaca, A., Jaber, N., Tsaran, V., and Kane, S. (2025). "Making Street View Accessible Using Context-Aware, Multimodal AI: A Demo of StreetReaderAI". Proceedings of *ASSETS '25*.
- [P.21] Muehlbradt, A., Koushik, V., and Kane, S.K. (2017). Goby: A wearable swimming aid for blind athletes. Proceedings of *ASSETS '17*, 2 pages.
- [P.20] Koushik, V., and Kane, S.K. (2017). Tangibles + programming + audio Stories = fun. Proceedings of *ASSETS '17*, 2 pages.
- [P.19] Kane, S.K., Profita, H., Lightner, M., and Correll, N. (2017). Assistive wearables: emerging trends and design considerations. Refereed presentation at CSUN '17 Conference.
- [P.18] Samson, S., Fiesler, C., and Kane, S.K. (2016). "Holy starches Batman!! We are getting walloped!": crowdsourcing comic book transcriptions. Proceedings of *ASSETS '16*, 289-290.

- [P.17] Lewis, C., Kane, S.K., and Ladner, R. (2016). Promoting strategic research on inclusive access to rich online content and services. *Proceedings of ASSETS '16*, 275-276.
- [P.16] Easley, W., Williams, M.A., Abdolrahmani, A., Galbraith, C., Branham, S.M., Hurst, A., and Kane, S.K. (2016). Let's get lost: exploring social norms in predominately blind environments. *Proceedings of CHI '16 Extended Abstracts*, 2034-2040.
- [P.15] Rode, J., Brady, E., Buehler, E., Kane, S.K., Ladner, R.E., Ringland, K.E., and Mankoff, J. 2016. SIG on the State of Accessibility at CHI. *Proceedings of CHI '16 Extended Abstracts*, 1100-1103.
- [P.14] Kane, S.K. and Bahram, S. (2015). Designing 3D-printed tactile comic books. Refereed presentation at CSUN '15 Conference.
- [P.13] Christian, C.A., Nota, A., Grice, N.A., Sabbi, E., Shaheen, N., Greenfield, P., Hurst, A., Kane, S., Rao, R., Dutterer, J., and de Mink, S.E. (2014). You can touch this! Bringing HST images to life as 3-D models. American Astronomical Society, AAS Meeting #223, id.244.16.
- [P.12] Calvo, R., Kane, S.K., and Hurst, A. (2014). Evaluating the accessibility of crowdsourcing tasks on Amazon's Mechanical Turk. *Proceedings of ASSETS '14*, 2 pages.
- [P.11] McDonald, S., Dutterer, J., Abdolrahmani, A., Kane, S.K., and Hurst, A. (2014). Tactile aids for visually impaired graphical design education. *Proceedings of ASSETS '14*, 2 pages.
- [P.10] Carrington, P., Hurst, A., and Kane, S.K. (2013). How power wheelchair users choose computing devices. *Proceedings of ASSETS '13*, 2 pages.
- [P.9] Brock, A.M., Kammoun, S., Nicolau, H., Guerreiro, T., Kane, S.K., and Jouffrais, C. (2013) SIG: NVI (Non-visual interaction). *Proceedings of CHI '13 Extended Abstracts*, 2513-2516.
- [P.8] Said, K. and Kane, S.K. (2013). Button Blender: remixing input to improve video game accessibility. *Proceedings of CHI '13 Extended Abstracts*, ACM, 43-48.
- [P.7] Zhu, S., Kane, S.K., Feng, J., and Sears, A. (2012). A crowdsourcing quality control model for tasks distributed in parallel. *Proceedings of CHI '12 Extended Abstracts*, 2501-2506.

- [P.6] Kane, S.K. (2009). Improving mobile phone accessibility with adaptive user interfaces. Proceedings of *iConference '09*. Poster.
- [P.5] Kane, S.K. and Klasnja, P.V. (2009). Supporting volunteer activities with mobile social software. Proceedings of *CHI '09 Extended Abstracts*, 4567-4572.
- [P.4] Cheung, G.C., Chilana, P.K., Kane, S.K. and Pellett, B. (2009). Designing for discovery: opening the hood for open-source end user tinkering. Proceedings of *CHI '09 Extended Abstracts*, 4321-4326.
- [P.3] Kane, S.K. and Wobbrock, J.O. (2007). Automatically correcting typing errors for people with motor impairments. Proceedings of *UIST '07 Extended Abstracts*, 2 pages.
- [P.2] Kane, S.K. (2007). Engaging student web programmers as inclusive designers. Proceedings of *ITICSE '07*. Poster.
- [P.1] Kane, S.K., Lehman, A. and Partridge, E. (2002). Indexing George Washington's handwritten manuscripts: a study of word matching techniques. *SIGCSE '02 Undergraduate Research Competition*. Poster.

Workshop Papers

- [W.6] Froehlich, J., Hwang, J., Wang, Z., O'Meara, J., Su, X., Huang, W., Zhang, Y., Fiannaca, A., Nelson, P., and Kane, S. (2025). “Does the cafe entrance look accessible? Where is the door?” Towards Geospatial AI Agents for Visual Inquiries”. Workshop on Vision Foundation Models and Generative AI for Accessibility at ICCV '25.
- [W.5] Taheri, A., Izadi, M., Shriram, G., Rostamzadeh, N., and Kane, S. (2023). “Breaking Barriers to Creative Expression: Co-Designing and Implementing an Accessible Text-to-Image Interface”. Workshop on Machine Learning for Creativity and Design at NeurIPS '23.
- [W.4] Kreiss, E., Venugopalan, S., Kane, S., and Morris, M.R. (2023). Practical challenges for investigating abbreviation strategies. CHI 2023 Workshop on Assistive Writing, 3 pages.
- [W.3] Fiannaca, A., Bennett, C., Kane, S., and Morris, M.R. (2022). Beyond Safety: toward a value-sensitive approach to the design of AI systems. NeurIPS 2022 Human-Centered AI Workshop, 6 pages.

[W.2] Hurst, A., and Kane, S.K. (2013). Making “making” accessible. Proceedings of *IDC '13*, 635-638.

[W.1] Williams, M.A., Hurst, A., and Kane, S.K. (2014). “Are you going to help her?” – ethical dilemmas presented by research with blind participants. Proceedings of *CSCW '14*, 4 pages.

Book Chapters

[BC.4] Kane, S.K. (2025). Mobile Development + Accessibility. In Alannah Oleson, Amy J. Ko, and Richard E. Ladner (Eds.). *Teaching Accessible Computing*.

[BC.3] Kane, S.K. (2019). Wearables. In Y. Yesilada and S. Harper (Eds.), *Web Accessibility*. Springer, 701-714.

[BC.2] Williams, M.A., Dubin, B., Amaefule, C., Nguyen, L., Abdolrahmani, A., Galbraith, C., Hurst, A., and Kane, S.K. (2016). Better supporting blind pedestrians and blind navigation technologies through accessible architecture. In P. Langdon, J. Lazar, A. Heylighen, and H. Dong (Eds.), *Designing Around People*. Springer, 237-246.

[BC.1] Kane, S.K., Hannah, J., Edwards, P.M., and Dorman, J. (2007). Teaching in computer classrooms. In C. Ross and J. Dunphy (Eds.), *Strategies for Teaching Assistant and International Teaching Assistant Development: Beyond Micro Teaching*. Bolton, MA: Anker, 48-52.

Invited Articles

[IA.4] Zimmermann-Niefield, A., Shapiro, R. B., and Kane, S.K. (2019). Sports and machine learning: how young people can use data from their own bodies to learn about machine learning. *XRDS: Crossroads, The ACM Magazine for Students*, 25(4), 44-49.

[IA.3] Kane, S.K. (2016). Invisibility, conspicuousness, and accessibility: how sensing systems fail for non-traditional users. Proceedings of *HCIC '16*, 3 pages.

[IA.2] Kane, S.K., Hurst, A., Buehler, E., Carrington, P.A., and Williams, M.A. (2014). Collaboratively designing assistive technology. *Interactions* 21(2), 78-81.

[IA.1] Kane, S.K. (2009). Context-enhanced interaction techniques for more accessible mobile phones. *SIGACCESS Newsletter, ACM*, 2 pages.

Invited Presentations

- [IP.16] Bigham, J.P., Lasecki, W., and Kane, S.K. (2017). AI for accessibility: augmenting sensory capabilities with intelligent technology. Microsoft Research Faculty Summit.
- [IP.15] Kane, S.K. (2017). Our assistive technologies, ourselves: understanding the relationships between wearable technology, accessibility, and identity. University of Texas Information School.
- [IP.14] Kane, S.K. (2016). Towards superhuman computing. Commencement speech, University of Washington Information School.
- [IP.13] Kane, S.K. (2016). Combining ability and context to create more accessible user interfaces. University of Washington DUB Seminar.
- [IP.12] Kane, S.K. (2016). Combining ability and context. Microsoft Research.
- [IP.11] Kane, S.K. (2015). Superhuman computing: designing custom software and hardware interfaces to support our natural abilities. Keynote talk at 2015 Richard Tapia Conference on Diversity and Computing.
- [IP.10] Kane, S.K. (2015). Superhuman computing: designing technology to support our natural abilities. University of Colorado ECSITE Program.
- [IP.9] Kane, S.K. (2013). Gesture-based user interfaces for a more accessible world. Human-Computer Interaction Institute, Carnegie Mellon University.
- [IP.8] Kane, S.K. (2013). Next generation accessible touch interfaces. Smith-Kettlewell Eye Research Institute.
- [IP.7] Kane, S.K. (2013). Mobile gesture-based user interfaces for people with disabilities. Microsoft Research.
- [IP.6] Kane, S.K. (2012). Gesture-based user interfaces for people with disabilities. IRIT (Toulouse, France).
- [IP.5] Kane, S.K. (2012). Prototyping and designing new assistive technologies for people with disabilities. Agency for Healthcare Research and Quality (AHRQ) Annual Conference.
- [IP.4] Kane, S.K. (2012). Understanding and creating usable touch interactions for blind people. HCIL Lab Talk, University of Maryland.

- [IP.3] Kane, S.K. and Shinohara, K. (2010) OneView: Enabling collaboration between blind and sighted students using tablet computers. Microsoft Research Faculty Summit.
- [IP.2] Kane, S.K. (2009). Supporting independent navigation using commodity mobile phones. Technology and Disability in the Developing World Conference, University of Washington.
- [IP.1] Kane, S.K. and Eisenberg, M.B. (2007). New tools for web credibility. Microsoft Research.

Technical Reports

- [T.6] Kane, S.K., Shinohara, K., and Wobbrock, J.O. (2015). *OneView: Enabling collaboration between blind and sighted students*. Technical report, University of Colorado.
- [T.5] Kane, S.K. and Galbraith, C. (2013). Design guidelines for creating voting technology for adults with aphasia. *Innovation Technology and Innovation Foundation Working Paper #006*.
- [T.4] Kane, S.K. (2005). Sketch-based input and evaluation in an online web-based learning environment. Master's thesis, University of Massachusetts.
- [T.3] Kane, S.K. (2003). Developing modular multi-user environments with Carnival. Undergraduate honors thesis, University of Massachusetts.
- [T.2] Rath, T.M., Kane, S.K., Lehman, A., Partridge, E. and Manmatha, R. (2002). Indexing for a digital library of George Washington's manuscripts: A study of word matching techniques. CIIR technical report MM-36, University of Massachusetts.
- [T.1] Kane, S.K., Lehman, A. and Partridge, E. (2001). Indexing George Washington's handwritten manuscripts. CIIR technical report MM-34, University of Massachusetts.

HONORS AND AWARDS

- SIGACCESS Paper Impact Award (2019, 2025) Received 2 awards given to “to the authors of an ASSETS conference paper that has had a significant impact on computing and information technology that addresses the needs of persons with disabilities”, for the papers Slide Rule: Making Mobile Touch Screens Accessible to Blind People Using Multi-Touch Interaction Techniques and Freedom to Roam: A Study of Mobile Device Adoption and Accessibility for People With Visual and Motor Disabilities
- ACM Distinguished Member (2023)
- JLS Paper of the Year Award (2023) For the paper *Reconfiguring science education through caring human inquiry and design with pets*.
- CU Boulder: Broadening Opportunity Through Leadership and Diversity (2018–2019)

Received fellowship from CU Boulder’s College of Engineering and Applied Science to explore ways to integrate universal design into engineering education.

- CU Boulder: Universal Design Fellowship (2016–2017)

Member of the inaugural cohort of CU Boulder Universal Design Fellows.

- University of Washington iSchool: Distinguished Alumni Award (2016)

Received Distinguished Alumni Award from the University of Washington Information School.

- UMass Amherst Computer Science: Outstanding Achievement Award (2016)

Received *Outstanding Achievement Award by a Young Alum* from the UMass Amherst College of Information and Computer Sciences.

- Sloan Research Fellowship (2016)

Two-year fellowship awarded yearly to early career researchers “in recognition of distinguished performance and a unique potential to make substantial contributions to their field.”

- UMBC Academic Innovation Fellowship (2014)

Awarded UMBC Academic Innovation Fellowship from the UMBC Hrabowski Innovation Fund for project *Exploring Opportunities and Challenges for Wearable Computing in Classroom Settings*.

- Google Lime Fellowship for Students with Disabilities (2010)

Fellowship provided to outstanding students with disabilities in the field of computing.

- Imagine Cup Touch and Tablet Accessibility Award, Finalist (2010)

Finalist for the Touch and Tablet Accessibility Award for the project *OneView: Enabling Collaboration between Blind and Sighted Students Using Tablet Computers*.

- NISH National Scholar Award for Workplace Innovation and Design, Hon. Mention (2009)

Received honorable mention for the NISH National Scholar Award for the submission *Fully Accessible Touch Screens for the Blind and Visually Impaired*.

- Most Innovative Award: UW Science and Engineering Business Association (2008)

Received Most Innovative Award at UW SEBA Science and Technology Showcase for project *Slide Rule: Eyes-Free Mobile Phone Applications for Everyone*.

- Graduate Student Top Scholar Award: University of Washington (2005)

Financial support provided to top applicants to the University of Washington Graduate School.

- Bay State Fellowship, University of Massachusetts (2003–2005)

Assistantship presented by the Department of Computer Science at the University of Massachusetts in recognition of undergraduate academic achievement.

- ACM SIGCSE Undergraduate Student Research Competition, Finalist (2002)

Awarded for the poster *Indexing George Washington's Handwritten Manuscripts: A Study of Word Matching Techniques*.

- National Science Foundation Research Experience for Undergraduates Fellowship (2001)

NSF REU research with R. Manmatha and James Allan at the Center for Intelligent Information Retrieval at the University of Massachusetts.

PAPER AWARDS

- Best Paper Award Nominee: ASSETS Conference (2020)

Received Best Paper Award nomination for the paper *Sense and Accessibility: Understanding people with physical disabilities' experiences with sensing systems*.

- Best Paper Honorable Mention: ASSETS Conference (2019) Received Best Paper Award nomination for the paper *RoboGraphics: Using Mobile Robots to Create Dynamic Tactile Graphics*.

- Honorable Mention: CHI Conference (2019)

Received Honorable Mention (top 5% of submissions) for the paper *StoryBlocks: A Tangible Programming Game to Create Accessible Audio Stories*.

- Best Paper Award: CHI Conference (2016)

Received Best Paper Award (top 1% of submissions) for the paper *Smart Touch: Improving Touch Accuracy for People with Motor Impairments with Template Matching*.

- Best Paper Award: CHI Conference (2015)

Received Best Paper Award (top 1% of submissions) for the paper *Sharing is Caring: Assistive Technology Designs on Thingiverse*.

- Best Paper Award: CHI Conference (2011)

Received Best Paper Award (top 1% of submissions) for the paper *Usable Gestures for Blind People: Understanding Preference and Performance*.

- Best Paper Award: MobileHCI Conference (2008)

Received Best Paper Award (top 2% of submissions) for the paper *Getting off the Treadmill: Evaluating Walking User Interfaces for Mobile Devices in Public Spaces*.

- Honorable Mention: CHI Conference (2017)

Received Honorable Mention (top 5% of submissions) for the paper *Toward Everyday Gaze Input: Accuracy and Precision of Eye Tracking and Implications for Design*.

- Honorable Mention: CHI Conference (2014)

Received Honorable Mention (top 5% of submissions) for the paper *Wearables and Chairables: Inclusive Design of Mobile Input and Output Devices for Power Wheelchair Users*.

- Best Workshop Paper: IDC Conference (2013)

Received Best Workshop Paper Award for the paper *Making “Making” Accessible*.

- Best Poster Award: iConference (2009)

Received Best Poster Award for the poster *Improving Mobile Phone Accessibility with Adaptive User Interfaces*.

- Outstanding Poster Award: ITICSE Conference (2007)

Received Outstanding Poster Award for the poster *Engaging Student Web Programmers as Inclusive Designers*.

RESEARCH FUNDING AND GRANTS

- CAREER: A New Interaction Model for Eyes-Free Exploration of Touch Screens

Funding source: National Science Foundation, Award #IIS-1652907

Project dates: 3/1/2017–2/28/2021

Investigator: Shaun Kane (PI) Award amount: \$546,610

- CHS: Small: User Interfaces for Improving Collaboration Between Blind and Sighted People

Funding source: National Science Foundation, Award #IIS-1619384

Project dates: 9/1/2016–8/31/2019

Investigator: Shaun Kane (PI) Award amount: \$499,997

- Sloan Research Fellowship

Funding source: Alfred P. Sloan Fellowship

Project dates: 3/1/2016–2/28/2018

Investigator: Shaun Kane (PI) Award amount: \$55,000

- CU Innovative Seed Grant: Increasing the Accessibility of Interactive Physics Simulations Using Actuated Tangible Interactions

Funding source: University of Colorado

Project dates: 3/1/2016–12/31/2017

Investigator: Shaun Kane (PI) Award amount: \$50,000

- Promoting Strategic Research on Inclusive Access to Rich Online Content and Services

Funding source: Computing Community Consortium

Project dates: 9/1/2015–12/31/2015

Investigators: Clayton Lewis (PI) and Shaun Kane (Co-PI) Award amount: \$16,000

- MRI: Acquisition of a 3D Object and Motion Capture System

Funding source: National Science Foundation, Award #CNS-1428204

Project dates: 9/1/2014–8/31/2017

Investigators: Marc Olano (PI), Shaun Kane (Co-PI); Amy Hurst, Daniel Bailey, Earle Ellis (Co-PIs) Award amount: \$175,195

- Exploring Opportunities and Challenges for Wearable Computing in Classroom Settings

Funding source: UMBC Hrabowski Innovation Fund

Project dates: 1/1/2014–12/31/2014

Investigator: Shaun Kane (PI) Award amount: \$14,000

- EAGER: Understanding Barriers to Workplace Collaboration for People with Visual Impairments

Funding source: National Science Foundation, Award #IIS-1353312

Project dates: 9/15/2013–8/31/2015

Investigator: Shaun Kane (PI) Award amount: \$164,982

- Exploring Ultra-Mobile Navigation Aids for Blind People

Funding source: Toyota Engineering and Manufacturing North America

Project dates: 1/1/2013–8/31/2014

Investigators: Amy Hurst and Shaun Kane (Joint-PI w/ equal responsibilities)
Award amount: \$380,753 (SK share 50%, \$22,399 to CU Boulder)

- Wheeltop Interaction: Full-Body Gesture Control for Power Wheelchair Users

Funding source: Microsoft Software Engineering and Innovation Foundation

Project dates: 9/1/2013–8/31/2014

Investigators: Amy Hurst and Shaun Kane (Joint-PI w/ equal responsibilities)
Award amount: \$25,000 (SK share 50%)

- Accessible Touch Interfaces for Power Wheelchair and Scooter Users

Funding source: Nokia University Cooperation Funding

Project dates: 9/1/2013–8/31/2014

Investigators: Amy Hurst and Shaun Kane (Joint-PI w/ equal responsibilities)
Award amount: \$11,450 (SK share 50%)

- Full-Body Gesture Control for Power Wheelchair Users

Funding source: UMBC Special Research Assistantship/Initiative Support

Project dates: 9/1/2013–8/31/2014

Investigators: Amy Hurst and Shaun Kane (Joint-PI w/ equal responsibilities)

Award amount: \$20,000 (SK share 50%)

- VotingVoice: A Mobile Voters' Guide for People with Aphasia

Funding source: Information Technology and Innovation Foundation

Project dates: 9/1/2012–8/31/2014

Investigator: Shaun Kane (PI) Award amount: \$102,434

- Audio-Haptic Feedback for Teaching Blind People to Use Touch Screens

Funding source: Google Research Award

Project dates: 9/1/2012–8/31/2013

Investigator: Shaun Kane (PI) Award amount: \$48,176

- Accessible Touch Screen-Based User Interfaces for Mobile Devices

Funding source: Foundation for Science and Disability

Project dates: 1/1/2010–12/31/2010

Investigator: Shaun Kane (PI) Award amount: \$1,000

TEACHING

Classes Taught

University of Colorado Boulder

- CSCI 4849/5849: Input, Interaction, and Accessibility (Spring 2018, Spring 2019, Fall 2020)
- CSCI 4830/7000: Inclusive Design and Assistive Technology (Spring 2017)
- CSCI 3002: Human-Centered Computing Foundations (Fall 2015, 2018, 2019, 2020)
- CSCI 4830/7000: Physical and Tangible Computing (Spring 2015)
- CSCI 5839: User-Centered Design and Development I (Fall 2014, 2016, 2017, 2019)

University of Maryland Baltimore County

- HCC 729: Human-Centered Design (Spring 2014)
- IS 403: User Interface Design (Fall 2013)
- IS 800: Mobile Human-Computer Interaction (Spring 2013)
- IS 760: Human-Computer Interaction (Fall 2011, Fall 2012)
- IS 387: Information Architecture for the World Wide Web (Spring 2012)

University of Washington

- INFO 463: Input and Interaction (Spring 2010, Spring 2011)
- INFO 344: Web Tools and Development (Summer 2006)

University of Massachusetts

- CS 121: Introduction to Problem Solving with Computers (Summer 2005)

Students Supervised

Ph.D. Advisees

- Darren Guinness, Ph.D. in Computer Science (2015–2022)
- Varsha Koushik, Ph.D. in Computer Science (2017–2022)
- Vinitha Gadiraju, Ph.D. in Computer Science (2018–2023)
- Halley P. Profita, co-advised with Nikolaus Correll, Ph.D. in Computer Science (2014–2017)
- Michele A. Williams, co-advised with Amy Hurst, Ph.D. in Human-Centered Computing at UMBC (2012–2015)

Ph.D. Supervisory Committee

- Junhan Kong, Ph.D. in Information Science, University of Washington (2025–)
- Mingyuan Zhong, Ph.D. in Information Science, University of Washington (2025–)
- Franklin Mingzhe Li, Ph.D. in Human-Computer Interaction, Carnegie Mellon University (2024–)

- Yichun Zhao, Ph.D. in Computer Science, University of Victoria (2024–)
- Guarav Jain, Ph.D. in Computer Science, Columbia University (2025–)
- Venkatesh Potluri, Ph.D. in Computer Science, University of Washington (2023–2024)
- Abir Saha, Ph.D. in Technology and Social Behavior, Northwestern University (2022–2024)
- Abigale Stangl, Ph.D. in ATLAS, University of Colorado Boulder (2015–2019)
- Redhwan Nour, Ph.D. in Computer Science, University of Colorado Boulder (2015–2017)
- Khalid Alharbi, Ph.D. in Computer Science, University of Colorado Boulder (2015–2016)
- Lise A. St. Denis, Ph.D. in Computer Science, University of Colorado Boulder (2014–2016)
- Markus Funk, Ph.D. in Human-Computer Interaction, University of Stuttgart (2016)
- Karl Wiegand, Ph.D. in Computer Science, Northeastern University (2012–2014)
- Huimin Qian, Ph.D. in Information Systems, UMBC (2011–2014)
- Shaojian Zhu, Ph.D. in Human-Centered Computing, UMBC (2012–2014)

Postdoctoral Fellows

- Stacy M. Branham, UMBC Information Systems (2013–2014). Now at UC Irvine.

Master's Thesis Advisees

- William Payne, M.S. in Computer Science, University of Colorado Boulder (2014–2016)
- Erin Duggan, co-advised with Tom Yeh, M.S. in Computer Science, University of Colorado Boulder (2014–2015)
- Ankita, M.S. in Computer Science, UMBC (2013–2014)

Master's Thesis Supervisory Committee

- Esther Vasiete, M.S. in Computer Science, University of Colorado Boulder (2014–2015)
- Jasmine Tobias, M.S. in Human-Centered Computing, UMBC (2013–2014)
- Ted O'Meara, M.S. in Human-Centered Computing, UMBC (2011)

Undergraduate Research Advisees

- Ariel Riggan, B.S. in Technology, Arts, and Media, University of Colorado. Discovery Learning Apprentice (2016–2017)
- Jesus Ortiz Tovar, B.S. in Computer Science, University of Colorado. Discovery Learning Apprentice (2016–2017)
- Christine Samson, B.S. in Computer Science, University of Colorado. Discovery Learning Apprentice (2015–2017)
- Sigrunn Sky, B.S. in Computer Science, University of Colorado. Research Assistant (2016)
- Laura Matuszewska, B.S. in Computer Science, University of Colorado. Discovery Learning Apprentice (2015)
- Lauren Gaber, B.S. in Computer Science, University of Michigan. AccessComputing Research Assistant (2015)
- Caroline Galbraith, B.S. in Information Systems, UMBC. Undergraduate Research Assistant (2012–2014)
- Beatrice Garcia, B.S. in Information Systems, UMBC. Undergraduate Research Assistant (2013–2014)
- Manpreet Suri, B.S. in Information Systems, UMBC. McNair Fellowship (2012)
- Kyle Althoff, B.S. in Information Systems, UMBC. Research Assistant (2012)
- Alec Pulianas, B.S. in Information Systems, UMBC. Research Assistant (2011–2012)
- Nicole Torcolini, Visiting Scholar, University of Washington. AccessComputing Research Assistant (2010)

- Tien Nguyen, M.S. in Information Management, University of Washington. Graduate Research Assistant (2010)
- Rishi Talwar, B.S. in Informatics, University of Washington. Directed Research Internship. (2010)
- Josh Scotland, B.S. in Computer Science, University of Washington. Undergraduate Research Assistant (2010)

SERVICE ACTIVITIES

Service to the Academic Community

- Conference Chair*
- General Chair, ASSETS Conference (2025–)

Journal Associate Editor

- ACM Transactions on Accessibility (2013–)
- International Journal of Human-Computer Studies (2011–2012)

Program Committees

- Program Co-Chair, ASSETS 2019 Conference
- Program Co-Chair, HCIC 2016 Conference
- Member, ACM Conference on Human Factors in Computing Systems (CHI 2012, 2013, 2015, 2017, 2018)
- Member, ACM Symposium on User Interface Software and Technology (UIST 2012, 2016, 2017)
- Member, ACM Conference on Computers and Accessibility (ASSETS 2012–2024)
- Member, ACM Web for All Conference (W4A 2015, 2018, 2019)
- Member, ACM Conference on Interaction Design and Children (IDC 2014)
- Member, ACM Conference on Intelligent User Interfaces (IUI 2009–2011)

Conference Committees

- Doctoral Consortium Co-Chair, ASSETS 2017

- Student Research Competition Judge, ASSETS 2012, ASSETS 2016
- Posters and Demonstrations Co-Chair, ASSETS 2015
- Accessibility Co-Chair, CHI 2016–2017
- Accessibility Co-Chair, CSCW 2014
- Doctoral School Panelist, ISWC 2014
- Doctoral Consortium Panelist, ASSETS 2012

Professional Organizations

- ACM SIGACCESS Member-at-Large (2016–2018)

Grant Referee

- National Science Foundation (2012, 2013, 2014, 2017, 2019)

Outreach

Advisory Boards

- Advisory board member, UW CREATE Center (2025–)
- Advisory board member, NYU Ability Project (2025–)
- Advisory board member, DIAGRAM Center, (2017–2022)
- AccessComputing partner (2014–)

Outreach Events

- Computing Research Association Underrepresented Minority and Disability Graduate Student Cohort Mentor (2018, 2019, 2023)
- Diverse Learners Awareness Week, University of Colorado Boulder (2017)
- Colorado Center for the Blind STEM Education Workshop (2014–2015)
- National Federation of the Blind Youth Slam (2011, 2013, 2017)